

PHYS 357: Honours Quantum Physics 1

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Abstract

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1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Taught by Bill Coish. Mondays, Wednesdays and Fridays from 9:35 to 10:25 in Steward Biology Building N2/2. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Stern-Gerlach Experiments

2.1 The classical picture

2.1.1 Angular momentum

Definition 1 (Orbital angular momentum). *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \quad (1)$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x - y plane, only the z component survives. On a circular path of radius r with speed v ,

$$L_z = mvr.$$

Definition 2 (Angular momentum of a rigid body). *For a rigid body¹ rotating about a fixed axis this is*

$$L = I_0\omega,$$

with I_0 the moment of inertia and ω the angular velocity.

Remark 1. *The moment of inertia I_0 measures how mass is distributed relative to the rotation axis.*

2.1.2 Magnetic moment of a circulating charge

Definition 3 (Magnetic moment). *A magnetic moment is a measure of the net circulation of electric charge. It is a vector quantity that determines the torque a magnet will experience in an external magnetic field.*

¹A rigid body is one that does not deform under the action of external forces.

The magnetic moment of a point particle of mass m and charge q in a circular orbit of radius r and period T is

$$\mu = \frac{IA}{c} = \frac{1}{c} \left(\frac{qv}{2\pi r} \right) (\pi r^2) = \frac{q}{2mc} mvr = \frac{q}{2mc} L_z,$$

where I is the current and A is the area enclosed by the orbit. A single circulating charge counts as a steady current $I = q/T = qv/2\pi r$ because it passes any given point on the loop once per period.

2.1.3 Gyromagnetic ratio

Definition 4 (Gyromagnetic ratio). *The gyromagnetic ratio γ is the constant of proportionality between the magnetic moment and the angular momentum,*

$$\vec{\mu} = \gamma \vec{L}. \quad (2)$$

For the circulating charge above,

$$\gamma = \frac{q}{2mc}. \quad (3)$$

Remark 2. *This is a purely classical result, since the charge is treated as a point particle on a definite orbit with a definite radius and speed, and nothing is quantized. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

2.2 Spin

Definition 5 (Spin). *Spin $\vec{S}$ is an intrinsic angular momentum carried by a particle. It is a fixed property of the particle type, like charge or mass, and does not come from any motion through space. A stationary electron alone in empty space still has spin.*

In quantum mechanics the spin $\vec{S}$ replaces the orbital angular momentum $\vec{L}$ of the classical picture.

2.2.1 Intrinsic moment and the g-factor

The magnetic moment is still proportional to the angular momentum, but the gyromagnetic ratio picks up an extra dimensionless factor,

$$\vec{\mu} = \gamma \vec{S}, \quad \gamma = \frac{qg}{2mc}, \quad (4)$$

where g is the g-factor of the particle.

Definition 6 (g-factor). *The g-factor of a particle³ is the dimensionless number relating its magnetic moment to its spin angular momentum. It measures how much stronger or weaker the moment is than the classical value $\gamma = q/2mc$, which corresponds to $g = 1$.*

²Here c is in Gaussian units.

³Its value must be measured or derived from relativistic quantum theory.

2.2.2 Magnetons

Definition 7 (Planck constant). *The Planck constant h is the fundamental constant of quantum mechanics. It sets the scale at which quantization matters, and relates the energy of a photon to its frequency by $E = h\nu$. Its value is*

$$h \approx 6.626 \times 10^{-34} \text{ J s.}$$

It has units of $J \cdot s$, which is the same as angular momentum.

Definition 8 (Reduced Planck constant).

$$\hbar = \frac{h}{2\pi}. \quad (5)$$

It carries units of angular momentum, so $[\hbar] = [L_z] = \frac{\text{kg} \cdot \text{m}^2}{\text{s}}$.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $q\hbar/2mc$ then carries the units and sets the natural size of the moment at a given mass scale.

Definition 9 (Bohr magneton). *The Bohr magneton is the natural unit of magnetic moment for an electron,*

$$\mu_B = \frac{|e|\hbar}{2m_e c}. \quad (6)$$

Definition 10 (Nuclear magneton). *The nuclear magneton is the same combination with the proton mass, and is the natural unit for nucleons,*

$$\mu_N = \frac{|e|\hbar}{2m_p c}. \quad (7)$$

Since $m_p/m_e \approx 1836$, we have $\mu_N \approx \mu_B/1836$.

Definition 11 (Nucleons). *The subatomic particles that make up the nucleus of an atom, specifically protons and neutrons.*

The electron moment is then written as

$$\vec{\mu}_e = g_e \mu_B \frac{\vec{S}}{\hbar}, \quad (8)$$

and similarly for the proton and neutron with μ_N in place of μ_B .

2.2.3 Values for the electron, proton and neutron

Particle	Charge q	Mass m	g -factor	Gyromagnetic ratio γ
Electron	$- e $	m_e	$g_e \approx 2$	$g_e \mu_B / \hbar$
Proton	$+ e $	m_p	$g_p \approx 5.59$	$g_p \mu_N / \hbar$
Neutron	0	m_p	$g_n \approx -3.83$	$g_n \mu_N / \hbar$

Table 1: Charges, masses, g -factors and gyromagnetic ratios.

Remark 3. The classical derivation gives $\mu = (q/2mc)L_z$, which vanishes when $q = 0$. The neutron is neutral yet has a nonzero moment, so its $\vec{\mu}$ cannot come from a single circulating point charge. This is a first sign that the classical picture fails.

Remark 4. Since $m_p \gg m_e$, we get $\mu_N \ll \mu_B$ and therefore $|\vec{\mu}_p| \ll |\vec{\mu}_e|$. Stern-Gerlach deflection of nuclei is much weaker than for electrons.

2.3 The Stern-Gerlach apparatus

Definition 12 (Stern-Gerlach apparatus). A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped to produce a strong field gradient along a chosen axis, here z . The gradient deflects each atom by an amount proportional to μ_z , and a screen records where it lands. Since $\vec{\mu} = \gamma\vec{S}$, the position on the screen is a direct measurement of S_z .

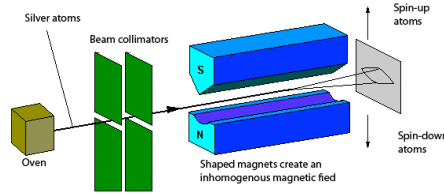


Figure 1: Illustration of the Stern-Gerlach apparatus.

2.3.1 Force in an inhomogeneous field

A magnetic moment in a field has potential energy

$$U = -\vec{\mu} \cdot \vec{B}, \quad (9)$$

so the force on it is

$$\vec{F} = -\vec{\nabla}U = \vec{\nabla}(\vec{\mu} \cdot \vec{B}), \quad F_z = -\frac{\partial U}{\partial z} = \mu_z \frac{\partial B_z}{\partial z}.$$

Definition 13 (Deflection). The displacement of an atom on the screen, perpendicular to the beam axis, produced by F_z acting over the length of the magnet. It is proportional to μ_z , so the pattern on the screen is a record of the moments in the beam.

Remark 5. A uniform field gives no force, only a torque. Thus, the apparatus needs a field gradient to deflect anything.

2.3.2 Silver atoms

Silver is used because its net orbital angular momentum vanishes, $\vec{L} = 0$, while $\vec{S} \neq 0$. The entire magnetic moment comes from a single unpaired valence electron, so the atom behaves as one electron spin. For the electron,

$$\mu_z = -\frac{g_e |e|}{2m_e c} S_z,$$

and therefore

$$F_z = \mu_z \frac{\partial B_z}{\partial z} = -g_e \frac{|e|}{2m_e c} S_z \frac{\partial B_z}{\partial z} = g_e \frac{|e|}{2m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z,$$

where the last equality assumes $\partial B_z / \partial z < 0$.

Result 1.

$$F_z \propto S_z.$$

The deflection of an atom measures the z component of its spin directly.

2.3.3 The classical prediction

Spin is a vector,

$$\vec{S} = (S_x, S_y, S_z)^T, \quad S_z = |\vec{S}| \cos \theta. \quad (10)$$

Remark 6. *The spins leaving the oven point in random directions, so θ takes every value in $[0, \pi]$ across the beam and $S_z = |\vec{S}| \cos \theta$ takes every value in $[-|\vec{S}|, |\vec{S}|]$. Since $F_z \propto S_z$, each atom is deflected differently and the screen should show a continuous smear, but it shows two spots.*

2.4 Quantization of S_z

Result 2 (Quantization of spin). *The electron spin angular momentum is quantized:*

$$S_z = \pm \frac{\hbar}{2}. \quad (11)$$

The beam splits into two spots.

2.4.1 Ket notation

Definition 14 (Basis state). *One of a set of states $\{|j\rangle\}$ that are mutually orthonormal and span the space, so that every state is a unique linear combination of them. For spin- $\frac{1}{2}$ the two outcomes of any one measurement, $\{|\pm \hat{n}\rangle\}$, form a basis.*

Definition 15 (Ket notation). A ket $|\cdot\rangle$ labels a quantum state as a vector in a vector space. For electron spin the two measurement outcomes give the basis states

$$|\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \quad |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2}.$$

The space is two-dimensional, so any spin state is a linear combination of the two.

Remark 7. From here on we follow Townsend and write $|+\hat{z}\rangle \equiv |\uparrow\rangle$ and $|-\hat{z}\rangle \equiv |\downarrow\rangle$. The same labels work for any axis: $|\pm\hat{x}\rangle$ are the states with $S_x = \pm\hbar/2$.

2.5 Sequential Stern-Gerlach experiments

Proposition 1. We write SG_z for a device whose gradient points along z , and SG_x for the same device rotated by 90° about the beam axis.

Definition 16 (Filter). Each device sends atoms into two beams. A filter is a device that blocks one of the beams.

Remark 8. A beam straight from the oven always splits 50/50 in any SG device, whatever its orientation.

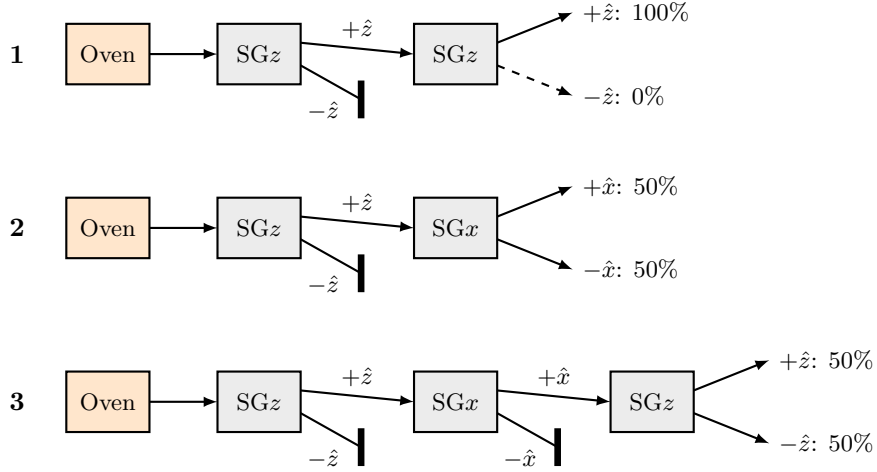


Figure 2: Sequential Stern-Gerlach experiments. Thick bars block a beam. Percentages are the fractions of atoms entering the last device. Every oven starts with N_0 atoms.

2.5.1 Experiment 1: SG_z then SG_z

Filter the $+\hat{z}$ beam and send it into a second SG_z . Every atom comes out $+\hat{z}$, none come out $-\hat{z}$. The measurement is repeatable: once an atom is found with $S_z = +\hbar/2$, measuring S_z again gives the same answer. The atom leaves the first device in the state $|+\hat{z}\rangle$. So we end with $N_0/2$ atoms with the $|+\hat{z}\rangle$.

2.5.2 Experiment 2: SG_z then SG_x

Filter the $+\hat{z}$ beam and send it into an SG_x. Half the atoms come out $+\hat{x}$ and half come out $-\hat{x}$. Knowing $S_z = +\hbar/2$ tells us nothing about S_x . We end up with $N_0/4$ in the $|+\hat{x}\rangle$ and $N_0/4$ in the $|-\hat{x}\rangle$.

2.5.3 Experiment 3: SG_z, SG_x, then SG_z

Filter $+\hat{z}$, then filter $+\hat{x}$, then measure S_z again. The beam splits 50/50 into $+\hat{z}$ and $-\hat{z}$ at the end, even though every atom was filtered to $+\hat{z}$ at the start. The S_x measurement wiped out the earlier S_z information. Blocking $+\hat{x}$ instead of $-\hat{x}$ gives the same 50/50 result. We end with $N_0/8|+\hat{z}\rangle$ and $N_0/8|-\hat{z}\rangle$.

Result 3. *An atom with a definite value of S_x does not have a definite value of S_z . Measuring S_x destroys what we knew about S_z , so no state has definite values of both at once.*

2.5.4 Experiment 4: SG_z, modified SG_x, then SG_z

Definition 17 (Modified SG device). *An SG device that splits the beam according to S and then recombines the two beams, so atoms with $S = \pm\hbar/2$ emerge at the same position even though they took two different paths.*

Filter $+\hat{z}$, send it through a modified SG_x with neither path blocked, then measure S_z . Every atom comes out $+\hat{z}$: 100%, not the 50/50 of Experiment 3. End with $N_0/4|+\hat{z}\rangle$ from $|+\hat{x}\rangle$ and $N_0/4|+\hat{z}\rangle$ from $|-\hat{x}\rangle$.

Result 4. *The modified SG_x cannot be sending $N_0/4$ through as $|+\hat{x}\rangle$ and $N_0/4$ as $|-\hat{x}\rangle$: Experiment 3 showed each of those alone gives 50/50 at the end. Instead the two paths interfere, as for water or light waves, cancelling the possibility of $-\hat{z}$. No path can be assigned to an atom.*

3 Quantum States as Vectors

3.1 The vector space of spin states

3.1.1 Superposition

Definition 18 (Superposition). *A general spin state is a linear combination of the basis kets,*

$$|\psi\rangle = c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle, \quad c_{\pm} \in \mathbb{C}. \quad (12)$$

The pair $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$ is a basis, and the coefficients $c_{\pm}$ are complex numbers.

3.1.2 Bras and the inner product

Definition 19 (Bra). *Every ket $|\psi\rangle$ has a matching bra $\langle\psi|$. Putting a bra in front of a ket gives a complex number, the inner product*

$$\langle\phi| |\psi\rangle \equiv \langle\phi|\psi\rangle \in \mathbb{C}.$$

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Definition 20 (Orthonormal basis). *The basis kets are orthonormal:*

$$\langle +\hat{z} | +\hat{z} \rangle = \langle -\hat{z} | -\hat{z} \rangle = 1, \quad \langle +\hat{z} | -\hat{z} \rangle = \langle -\hat{z} | +\hat{z} \rangle = 0. \quad (13)$$

3.1.3 Completeness

Remark 9. *The Born rule, $P(\phi | \psi) = |\langle \phi | \psi \rangle|^2$, is the probability of finding $|\phi\rangle$ when $|\psi\rangle$ is prepared. Why is it true? There is no deeper derivation: it is a postulate that agrees with every experiment. As David Mermin put it, "shut up and calculate."*

Definition 21 (Projector). *The operator $P_j = |j\rangle \langle j|$ built from a basis state. Acting on $|\psi\rangle = \sum_k c_k |k\rangle$ it keeps only the $|j\rangle$ part,*

$$P_j |\psi\rangle = c_j |j\rangle,$$

since $\langle j | k \rangle = \delta_{jk}$.

Proposition 2 (Completeness relation). *Since any state can be expanded in the basis $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$, the sum of the projectors onto the basis states is the identity:*

$$|+\hat{z}\rangle \langle +\hat{z}| + |-\hat{z}\rangle \langle -\hat{z}| = 1. \quad (14)$$

Applying this to $|\hat{x}\rangle$ gives the expansion directly:

$$|\hat{x}\rangle = (|+\hat{z}\rangle \langle +\hat{z}| + |-\hat{z}\rangle \langle -\hat{z}|) |\hat{x}\rangle = \langle +\hat{z} | \hat{x} \rangle |+\hat{z}\rangle + \langle -\hat{z} | \hat{x} \rangle |-\hat{z}\rangle, \quad (15)$$

3.2 Extracting probabilities

3.2.1 Probability amplitudes

Proposition 3. *To pull out c_+ , act on $|\psi\rangle$ with $\langle +\hat{z}|$:*

$$\langle +\hat{z} | \psi \rangle = c_+ \underbrace{\langle +\hat{z} | +\hat{z} \rangle}_{=1} + c_- \underbrace{\langle +\hat{z} | -\hat{z} \rangle}_{=0} = c_+. \quad (16)$$

The same steps with $\langle -\hat{z}|$ give c_- . So

$$c_{\pm} = \langle \pm\hat{z} | \psi \rangle. \quad (17)$$

Definition 22 (Probability amplitude). *The coefficient $c_{\pm} = \langle \pm\hat{z} | \psi \rangle$ is the probability amplitude for measuring $S_z = \pm\hbar/2$. The probability itself is the modulus squared,*

$$P(S_z = \pm\hbar/2) = |c_{\pm}|^2 = |\langle \pm\hat{z} | \psi \rangle|^2. \quad (18)$$

Principle 1 (Born rule). *The probability that a system in the state $|\psi\rangle$ is found in the state $|\phi\rangle$ is*

$$P(\phi | \psi) = |\langle \phi | \psi \rangle|^2 \in [0, 1]. \quad (19)$$

3.2.2 Normalization

Proposition 4. *A system in $|\psi\rangle$ is found in $|\psi\rangle$ with certainty, so*

$$P(\psi|\psi) = |\langle\psi|\psi\rangle|^2 = 1. \quad (20)$$

To find what a bra equals to, write it with unknown coefficients, $\langle\psi| = c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|$. Then

$$\langle\psi|\psi\rangle = (c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|) (c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle) = c'_+ c_+ + c'_- c_-,$$

since the cross terms vanish by orthonormality. The two outcomes $S_z = \pm\hbar/2$ must have probabilities $|c_+|^2$ and $|c_-|^2$ that add to 1. For $\langle\psi|\psi\rangle$ to be that sum, each product $c'_\pm c_\pm$ must be real and equal to $|c_\pm|^2$. This forces $c'_\pm = c_\pm^*$.

Result 5. *Going from a ket to its bra, conjugate the coefficients:*

$$\langle\psi| = c_+^* \langle+\hat{z}| + c_-^* \langle-\hat{z}|. \quad (21)$$

In particular $\langle\psi|\pm\hat{z}\rangle = c_\pm^ = \langle\pm\hat{z}|\psi\rangle^*$. More generally, $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$.*

Definition 23 (Normalization condition).

$$\langle\psi|\psi\rangle = \langle\psi|+\hat{z}\rangle \langle+\hat{z}|\psi\rangle + \langle\psi|-\hat{z}\rangle \langle-\hat{z}|\psi\rangle = |c_+|^2 + |c_-|^2 = 1. \quad (22)$$

In words, $P(S_z = +\hbar/2) + P(S_z = -\hbar/2) = 1$: the atom is always found in one of the two beams.

Example. *Take*

$$|\psi\rangle = \frac{1}{\sqrt{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle.$$

The amplitude for $S_z = +\hbar/2$ is

$$\langle+\hat{z}|\psi\rangle = \frac{1}{\sqrt{3}} \underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + i\sqrt{\frac{2}{3}} \underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = \frac{1}{\sqrt{3}},$$

so $P(S_z = +\hbar/2) = 1/3$. In the same way $\langle-\hat{z}|\psi\rangle = i\sqrt{2/3}$, so $P(S_z = -\hbar/2) = |i|^2 \cdot 2/3 = 2/3$. The two add to 1, so the state is normalized.

3.2.3 Global phase

Proposition 5. *Take two kets that differ only by an overall phase, $|\psi\rangle = e^{i\alpha} |\psi'\rangle$ with $\alpha \in \mathbb{R}$. For any state $|\phi\rangle$,*

$$P(\phi|\psi) = |\langle\phi|\psi\rangle|^2 = |e^{i\alpha} \langle\phi|\psi'\rangle|^2 = |\langle\phi|\psi'\rangle|^2 = P(\phi|\psi'), \quad (23)$$

since $|e^{i\alpha}| = 1$.

Result 6. $|\psi\rangle$ and $e^{i\alpha}|\psi\rangle$ give the same probabilities for every possible measurement, so they describe the same physical state. An overall (global) phase has no physical meaning and can be dropped.

Dropping the global phase $e^{i\delta_+}$ from $|+\hat{x}\rangle$ leaves

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle).$$

Remark 10. The relative phase δ is different: it sits between the two terms, so it cannot be pulled out front and dropped. Experiment 3 alone does not fix it. It is pinned down by combining more measurements (for example along y) with a phase convention, and the standard choice for $|+\hat{x}\rangle$ is $\delta = 0$.

3.3 Fixing the Basis States

3.3.1 The state $|+\hat{x}\rangle$

In Experiment 3, atoms in $|+\hat{x}\rangle$ enter an SG z and split 50/50:

$$P(S_z = \pm \frac{\hbar}{2}) = |\langle \pm \hat{z} | +\hat{x} \rangle|^2 = \frac{1}{2}.$$

So both amplitudes have modulus $1/\sqrt{2}$, and each can carry its own phase:

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} e^{i\delta_+} |+\hat{z}\rangle + \frac{1}{\sqrt{2}} e^{i\delta_-} |-\hat{z}\rangle = \frac{e^{i\delta_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle), \quad \delta = \delta_- - \delta_+,$$

with $\delta_{\pm} \in \mathbb{R}$.

3.3.2 Experiment 5: measuring S_y

Definition 24 (Cyclic permutation). A cyclic permutation of the axes relabels them as $x \rightarrow y$, $y \rightarrow z$, $z \rightarrow x$ (or the reverse). It is a rotation of the coordinate system, so no physics changes under it.

Let Experiment 5 be a cyclic permutation of Experiment 3, SG x becomes SG z and SG z becomes SG y . Now the process is as follows: filter $+\hat{z}$, then measure S_y .

Proposition 6. By symmetry it must split 50/50 as well, so every cross-axis overlap has the same modulus:

$$|\langle +\hat{x} | +\hat{z} \rangle|^2 = |\langle +\hat{y} | +\hat{z} \rangle|^2 = |\langle -\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2}. \quad (24)$$

The same argument as for $|+\hat{x}\rangle$ then gives

$$|+\hat{y}\rangle = \frac{e^{i\gamma_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\gamma} |-\hat{z}\rangle), \quad \gamma = \gamma_- - \gamma_+.$$

3.3.3 The state $|+\hat{y}\rangle$

Proposition 7. *Drop the global phases, $\gamma_+ = \delta_+ = 0$, so that we finally obtain equations for x and y in terms of z :*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\delta} |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\gamma} |-\hat{z}\rangle).$$

So using orthonormality of the $\hat{z}$ basis,

$$\langle +\hat{x} | +\hat{y} \rangle = \frac{1}{2} (1 + e^{i(\gamma-\delta)}), \quad |\langle +\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2} (1 + \cos(\gamma - \delta)).$$

Requiring this to equal $1/2$ forces $\cos(\gamma - \delta) = 0$, that is $\gamma - \delta = \pm\pi/2$. The remaining freedom is convention.

Result 7 (Phase convention). *Take $\delta = 0$ and $\gamma = +\pi/2$. Then*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm i |-\hat{z}\rangle), \quad (25)$$

The bras are then

$$\langle \pm\hat{x} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \pm \langle -\hat{z} |), \quad \langle \pm\hat{y} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \mp i \langle -\hat{z} |). \quad (26)$$

Remark 11. *Complex coefficients are required, not just allowed. With real coefficients the relative phases could only be 0 or π , so $\gamma - \delta = \pm\pi/2$ would be impossible.*

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle$, compute $\langle S_y \rangle$.*

The amplitudes are

$$\begin{aligned} \langle +\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle +\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle +\hat{y} | -\hat{z} \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}} (-i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + \sqrt{\frac{2}{3}} \right), \\ \langle -\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle -\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle -\hat{y} | -\hat{z} \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}} (i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} - \sqrt{\frac{2}{3}} \right). \end{aligned}$$

Computing the probabilities

$$P(S_y = +\frac{\hbar}{2}) = |\langle +\hat{y} | \psi \rangle|^2 = \frac{1}{2} + \frac{\sqrt{2}}{3}, \quad P(S_y = -\frac{\hbar}{2}) = |\langle -\hat{y} | \psi \rangle|^2 = \frac{1}{2} - \frac{\sqrt{2}}{3},$$

which add to 1 as they must. Then

$$\langle S_y \rangle = \frac{\hbar}{2} \left[\left(\frac{1}{2} + \frac{\sqrt{2}}{3} \right) - \left(\frac{1}{2} - \frac{\sqrt{2}}{3} \right) \right] = \frac{\sqrt{2}}{3} \hbar.$$

The factor i that did nothing for S_z matters here: it is a relative phase, and it is exactly what tilts the spin toward $+\hat{y}$.

3.4 Statistics of Measurements

3.4.1 Expectation value

Definition 25 (Expectation value). *The expectation value of S_z in the state $|\psi\rangle$ is the average outcome weighted by the probabilities,*

$$\langle S_z \rangle = \left(\frac{\hbar}{2}\right) |\langle +\hat{z}|\psi \rangle|^2 + \left(-\frac{\hbar}{2}\right) |\langle -\hat{z}|\psi \rangle|^2. \quad (27)$$

Remark 12. *For N measurements on identically prepared atoms, with N_+ giving $+\hbar/2$ and N_- giving $-\hbar/2$, the measured average is*

$$\langle S_z \rangle_N = \frac{\hbar}{2} \frac{N_+ - N_-}{N}, \quad (28)$$

which tends to $\langle S_z \rangle$ as $N \rightarrow \infty$.

Remark 13 (Shot noise). *For a finite number N of measurements, the sample average deviates from the expectation value by*

$$|\langle S_z \rangle_N - \langle S_z \rangle| \sim \frac{\Delta S_z}{\sqrt{N}}. \quad (29)$$

This spread is intrinsic, not a flaw of the apparatus: it appears whenever a state with a definite value of one observable (e.g. S_x for $|+\hat{x}\rangle$) is measured for an incompatible observable (e.g. S_z).

3.4.2 Uncertainty

Definition 26 (Uncertainty). *The uncertainty ΔS_z is the standard deviation of the measurement outcomes,*

$$(\Delta S_z)^2 = \langle S_z^2 \rangle - \langle S_z \rangle^2, \quad (30)$$

where

$$\langle S_z^2 \rangle = \left(\frac{\hbar}{2}\right)^2 |\langle +\hat{z}|\psi \rangle|^2 + \left(-\frac{\hbar}{2}\right)^2 |\langle -\hat{z}|\psi \rangle|^2. \quad (31)$$

Example. *For $|+\hat{x}\rangle$, both probabilities are $1/2$, so*

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{2} - \frac{\hbar}{2} \cdot \frac{1}{2} = 0, \quad \langle S_z^2 \rangle = \left(\frac{\hbar}{2}\right)^2 \cdot \frac{1}{2} + \left(-\frac{\hbar}{2}\right)^2 \cdot \frac{1}{2} = \frac{\hbar^2}{4},$$

and therefore $\Delta S_z = \hbar/2$. The average is zero but every single measurement gives $\pm\hbar/2$, so the spread is maximal.

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}}|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$, the probabilities are $1/3$ and $2/3$, so*

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{3} - \frac{\hbar}{2} \cdot \frac{2}{3} = -\frac{\hbar}{6}, \quad \langle S_z^2 \rangle = \frac{\hbar^2}{4} \left(\frac{1}{3} + \frac{2}{3} \right) = \frac{\hbar^2}{4}.$$

Then

$$(\Delta S_z)^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{36} = \frac{9\hbar^2 - \hbar^2}{36} = \frac{2}{9}\hbar^2, \quad \Delta S_z = \frac{\sqrt{2}}{3}\hbar.$$

Remark 14. *$\langle S_z^2 \rangle$ is always equal to $\frac{\hbar^2}{4}$.*

3.5 Generalization to n -level systems

3.5.1 Orthonormality and completeness

A spin- $\frac{1}{2}$ particle is a two-level system, and each of $\{|\pm\hat{x}\rangle\}$, $\{|\pm\hat{y}\rangle\}$, $\{|\pm\hat{z}\rangle\}$ is a basis. An n -level system has a basis $\{|j\rangle\}$, $j = 1, \dots, n$.

Remark 15. *The index j labels the outcomes of one chosen observable, not the coordinate axes. Picking S_x , S_y or S_z picks the basis; within that basis $j = 1, 2$ labels its two states $|a_1\rangle = |+\hat{x}\rangle$ and $|a_2\rangle = |-\hat{x}\rangle$. A spin-1 particle has $n = 3$ outcomes, and an n -level atom has $j = 1, \dots, n$.*

Definition 27 (Spin quantum number). *The spin quantum number s labels the magnitude of a particle's spin and is fixed by the particle type. It takes the values $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$, and a measurement of S_z then gives one of $2s + 1$ outcomes,*

$$S_z = m\hbar, \quad m = -s, -s + 1, \dots, +s. \quad (32)$$

The electron has $s = \frac{1}{2}$, so $S_z = \pm\hbar/2$ and the beam splits in two. A spin-1 particle would split into three beams, $S_z = +\hbar, 0, -\hbar$.

Definition 28 (Orthonormal and complete basis). *The basis is orthonormal if*

$$\langle i|j\rangle = \delta_{ij},$$

and complete if every state can be expanded in it,

$$|\psi\rangle = \sum_j c_j |j\rangle, \quad c_j = \langle j|\psi\rangle, \quad \langle\psi| = \sum_j c_j^* \langle j|. \quad (33)$$

Completeness is equivalent to

$$\mathbb{1} = \sum_{j=1}^n |j\rangle \langle j|. \quad (34)$$

Example. *For $n = 2$, $\mathbb{1} = |+\hat{n}\rangle \langle +\hat{n}| + |-\hat{n}\rangle \langle -\hat{n}|$ for any $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$, so*

$$|\psi\rangle = \langle +\hat{x}|\psi\rangle |+\hat{x}\rangle + \langle -\hat{x}|\psi\rangle |-\hat{x}\rangle = \langle +\hat{y}|\psi\rangle |+\hat{y}\rangle + \langle -\hat{y}|\psi\rangle |-\hat{y}\rangle.$$

3.5.2 Observables and the Born rule

Definition 29 (Observable). *An observable A has outcomes a_j with states $|a_j\rangle$ forming a complete orthonormal set. For spin- $\frac{1}{2}$, A could be S_x , S_y or S_z , with outcomes $\pm\hbar/2$. By the Born rule,*

$$P(A = a_j) = |\langle a_j|\psi\rangle|^2. \quad (35)$$

Proposition 8. *Normalization $\langle\psi|\psi\rangle = \mathbb{1}$ is the statement that the probabilities sum to 1.*

Proof.

$$\langle \psi | \psi \rangle = \left(\sum_i c_i^* \langle a_i | \right) \left(\sum_j c_j | a_j \rangle \right) = \sum_{i,j} c_i^* c_j \delta_{ij} = \sum_j |c_j|^2 = \sum_j P(A = a_j) = 1.$$

Equivalently, inserting the identity, $\sum_j \langle \psi | a_j \rangle \langle a_j | \psi \rangle = \langle \psi | \left(\sum_j | a_j \rangle \langle a_j | \right) | \psi \rangle = \langle \psi | \psi \rangle$. $\square$

3.5.3 Expectation value as an operator

Proposition 9.

$$\langle A \rangle = \sum_j a_j |\langle a_j | \psi \rangle|^2 = \sum_j \langle \psi | a_j \rangle a_j \langle a_j | \psi \rangle = \langle \psi | \hat{A} | \psi \rangle, \quad \hat{A} = \sum_j a_j | a_j \rangle \langle a_j |. \quad (36)$$

Definition 30 (Uncertainty of an observable).

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2, \quad \langle A^2 \rangle = \langle \psi | \hat{A}^2 | \psi \rangle = \sum_j a_j^2 |c_j|^2. \quad (37)$$

4 Rotation of Basis States and Matrix Mechanics

4.1 States as column vectors

Definition 31 (Representation of a state). *In the S_z basis, a ket is represented by the column of its components and a bra by the conjugate row, the adjoint:*

$$|\psi\rangle \xrightarrow{S_z} \psi = \begin{pmatrix} c_+ \\ c_- \end{pmatrix}, \quad \langle \psi | \xrightarrow{S_z} \psi^\dagger = (c_+^* \quad c_-^*), \quad (38)$$

with $c_\pm = \langle \pm \hat{z} | \psi \rangle$. Then

$$\langle \psi | \psi \rangle = \psi^\dagger \psi = |c_+|^2 + |c_-|^2. \quad (39)$$

Remark 16. Write $|\psi\rangle \xrightarrow{S_z} \psi$, not $|\psi\rangle = \psi$. The ket is basis-independent; the column depends on the basis, just like $\vec{E}$ versus its components.

Example.

$$|\pm \hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \quad |\pm \hat{y}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

Example (Other bases). In the S_x basis, $|\psi\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x} | \psi \rangle \\ \langle -\hat{x} | \psi \rangle \end{pmatrix}$, and similarly for S_y . So the same state has different columns:

$$|+\hat{z}\rangle \xrightarrow{S_z} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |+\hat{z}\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x} | +\hat{z} \rangle \\ \langle -\hat{x} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |+\hat{z}\rangle \xrightarrow{S_y} \begin{pmatrix} \langle +\hat{y} | +\hat{z} \rangle \\ \langle -\hat{y} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

4.2 Operators as matrices

Definition 32 (Projection operator). $P_{\pm\hat{n}} = |\pm\hat{n}\rangle\langle\pm\hat{n}|$, so that $\mathbb{1} = P_{+\hat{n}} + P_{-\hat{n}}$ and $P_{+\hat{z}}|\psi\rangle = c_+|+\hat{z}\rangle$.

Definition 33 (Matrix of an operator). In a basis $\{|j\rangle\}$, an operator A is represented by the matrix

$$A_{ij} = \langle i|A|j\rangle. \quad (40)$$

Inserting the identity, $\langle k|A|\psi\rangle = \sum_j \langle k|A|j\rangle\langle j|\psi\rangle$, so $A|\psi\rangle$ is represented by the matrix product $A\psi$.

Example.

$$P_{+\hat{z}} \xrightarrow{S_z} \begin{pmatrix} \langle +\hat{z}|P_{+\hat{z}}|+\hat{z}\rangle & \langle +\hat{z}|P_{+\hat{z}}|-\hat{z}\rangle \\ \langle -\hat{z}|P_{+\hat{z}}|+\hat{z}\rangle & \langle -\hat{z}|P_{+\hat{z}}|-\hat{z}\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} c_+ \\ c_- \end{pmatrix} = \begin{pmatrix} c_+ \\ 0 \end{pmatrix}.$$

Proposition 10 (Properties of projectors). A projector satisfies $P_{\hat{n}}^2 = P_{\hat{n}}$, since $(|\hat{n}\rangle\langle\hat{n}|)(|\hat{n}\rangle\langle\hat{n}|) = |\hat{n}\rangle\langle\hat{n}|\hat{n}\rangle\langle\hat{n}| = |\hat{n}\rangle\langle\hat{n}|$. In the S_z basis,

$$P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbb{1} = P_{+\hat{z}} + P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (41)$$

Definition 34 (Adjoint). Writing $A = \mathbb{1}A\mathbb{1} = \sum_{i,j} A_{ij} |i\rangle\langle j|$, the adjoint is

$$A^\dagger = \sum_{i,j} A_{ij}^* |j\rangle\langle i|, \quad (A^\dagger)_{ij} = A_{ji}^*. \quad (42)$$

If $|\psi'\rangle = A|\psi\rangle$, then $\langle\psi'| = \langle\psi|A^\dagger$.

4.3 The rotation operator

Definition 35 (Rotation operator). $R(\theta\hat{n})$ rotates a state by θ about $\hat{n}$: $|\psi'\rangle = R(\theta\hat{n})|\psi\rangle$. For example, $R(\frac{\pi}{2}\hat{y})$ takes $|\pm\hat{z}\rangle$ to $|\pm\hat{x}\rangle$ (up to phase), so

$$R\left(\frac{\pi}{2}\hat{y}\right)(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+|+\hat{x}\rangle + c_-|-\hat{x}\rangle. \quad (43)$$

Remark 17. $R(\theta\hat{n})$ is a linear operator: $R(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+R|+\hat{z}\rangle + c_-R|-\hat{z}\rangle$, which is what the equation above uses.

Remark 18. The bra of $|\psi'\rangle$ is $\langle\psi|R^\dagger$, not $\langle\psi|R$. With the wrong choice, $|\psi\rangle = |+\hat{z}\rangle$ and $\theta = \pi/2$ would give $\langle\psi'|\psi'\rangle = \langle+\hat{z}|R(\pi\hat{y})|+\hat{z}\rangle = \langle+\hat{z}|-\hat{z}\rangle = 0$ instead of 1.

Theorem 1. Rotation operators are unitary:

$$R^\dagger R = R R^\dagger = \mathbb{1}. \quad (44)$$

Proof. A rotation must preserve normalization: $\langle\psi'|\psi'\rangle = \langle\psi|R^\dagger R|\psi\rangle = 1$ for every $|\psi\rangle$ and every $R(\theta\hat{n})$, which forces $R^\dagger R = \mathbb{1}$. $\square$

Remark 19. Unitary operators map physical (normalized) states to physical states. If a unitary U satisfies $U|\psi\rangle = c|\psi\rangle$, then $|c| = 1$, so $c = e^{i\varphi}$ is just a global phase and $U|\psi\rangle$ describes the same physical state as $|\psi\rangle$.

A Appendix

B Useful Links

Stern-Gerlach experiment: <https://www.youtube.com/shorts/QG7kJXxN1Kk>